Calendars and Time-Keeping across Cultures

Covers Lectures 1.4–1.6 about 90 minutes

Name: _____ Date: _____ Marks: _____ / 80

You may use: a basic calculator and your lecture notes (try Sections A–C from memory first). **Difficulty:** ★ remember ★★ apply ★★★ challenge

Section A — Fill in the blanks ★ 12 marks

1. The five limbs of the Panchang are tithi, vara, _____, yoga and karana.
2. A tithi ends each time the Moon gains _____° on the Sun.
3. There are _____ tithis in one lunar month.
4. The waxing (bright) fortnight is called _____ paksha.
5. The last tithi of the dark fortnight is _____.
6. An Adhik Maas is added about _____ times in every 19 years.
7. 1 ghati = _____ minutes and 1 muhurta = _____ minutes.
8. The Islamic calendar counts years from the _____, which happened in _____ CE.
9. Ramadan is the _____th Islamic month, and Eid al-Fitr falls on the 1st of _____.
10. An Islamic day begins at _____.
11. The Gregorian calendar was introduced by Pope _____ in the year _____.
12. Indian Standard Time is based on the _____° E meridian and equals UTC + _____.

Section B — Multiple choice ★ 10 marks

1. Which is **NOT** one of the five limbs of the Panchang? (a) tithi (b) nakshatra (c) rashi (d) karana
2. The average length of a tithi is about: (a) 24 h 00 m (b) 23 h 37 m (c) 25 h 10 m (d) 22 h 00 m
3. The month in which the Full Moon occurs near the **Pleiades (Krittika)** is: (a) Chaitra (b) Kartika (c) Magha (d) Pausha
4. An Adhik Maas is added mainly to: (a) keep lunar months in step with the seasons (b) correct for the Moon slowing down (c) add leap seconds (d) make the weeks fit
5. A Hijri year has: (a) 365 days (b) 354 or 355 days (c) 360 days (d) 384 days
6. Which year was/will be **NOT** a leap year? (a) 2000 (b) 2024 (c) 1900 (d) 2028
7. When British India adopted the Gregorian calendar in 1752, it dropped: (a) 10 (b) 11 (c) 13 (d) 12 days
8. The month name "September" originally meant: (a) ninth month (b) seventh month (c) tenth month (d) month of summer
9. The Indian National Calendar (1957) was recommended by a committee chaired by: (a) C. V. Raman (b) Meghnad Saha (c) Homi Bhabha (d) Vikram Sarabhai
10. "Guruvar" and "Thursday" are both named after: (a) Mars (b) Mercury (c) Jupiter (d) Saturn

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Section C — Memory patterns ★/★★ 20 marks

C1. The weekday star (4 marks). Write the grahas in Chaldean order (slowest → fastest), then, starting from the Sun, jump 3 places each time to generate the weekdays.

Chaldean order	1	2	3	4 Sun	5	6	7
Planet							

Weekday	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Planet	Sun						
Sanskrit/Hindi name	Ravivar						

C2. Tithi sequence (3 marks). Fill in the gaps: Pratipada, _____, Tritiya, _____, Panchami, _____, Saptami, _____, Navami, Dashami, _____, Dwadashi, _____, Chaturdashi, Purnima / _____.

C3. Months and nakshatras (3 marks).

Lunar month	Full Moon near nakshatra (star)
Chaitra	
	Krittika (Pleiades)
Pausha	
Magha	
	Shravana (Altair)
Jyeshtha	

C4. Islamic months in order (3 marks). Number them 1–12 by position in the year: Ramadan _____, Muharram _____, Shawwal _____, Dhu al-Hijjah _____, Rabi' al-Awwal _____, Rajab _____.

C5. Festival match (4 marks, ½ each).

Festival	Calendar date
1. Holi	A. 10 Dhu al-Hijjah
2. Diwali (Lakshmi Puja)	B. Bhadrapada Shukla Chaturthi
3. Ganesh Chaturthi	C. 1 Shawwal
4. Guru Purnima	D. Phalguna Purnima
5. Raksha Bandhan	E. 10 Muharram
6. Eid al-Adha	F. Amavasya of Ashvin/Kartika
7. Eid al-Fitr	G. Ashadha Purnima
8. Ashura	H. Shravana Purnima

C6. Roman months (3 marks).

Month	Original Latin meaning	Original position	Sanskrit number word
September			sapta (7)
October			
November			
December			

Section D — Short answers ★★ 10 marks

1. Why does the Gregorian date of **Diwali** or **Holi** change by up to three weeks from year to year, while **Makar Sankranti** stays on 14–15 January?
2. Why may **Eid** be celebrated on different days in India and in Saudi Arabia?
3. Explain **Amanta** and **Purnimanta** months using the example of Janmashtami.
4. Why did Pope Gregory's reform remove **10 days** in 1582? Was the order of the weekdays disturbed?
5. Hindu and Islamic calendars both use lunar months. Why do Hindu festivals stay in roughly the same season while Islamic festivals move through all seasons?

Section E — Calculations ★★ 28 marks

Formula box

Synodic month = 29.53 d · tropical year = 365.24 d · 12 lunar months = 354.37 d
Tithi number = $\lceil E/12° \rceil$ (16–30 → Krishna; subtract 15) · 1 pala = 24 s · 1 ghati = 24 min · 1 muhurta = 48 min · 1 prahar = 3 h
$\text{AH} \approx (\text{CE} - 622) \times \frac{33}{32}$ $\text{CE} \approx \text{AH} \times \frac{32}{33} + 622$ Local mean time = IST + 4 min × $(\lambda - 82.5°)$

Worked example

The Moon is 140° ahead of the Sun: $140/12 = 11.67 \Rightarrow \lceil 11.67 \rceil = 12 \Rightarrow$ **Shukla Dwadashi**.

E1. [5] Name the tithi (number, paksha, name) for an elongation of (a) 30° (b) 95° (c) 185° (d) 335° (e) 359°.

E2. [3] On a Wednesday the 1st hora is ruled by Mercury. (a) Which planets rule the 2nd, 3rd and 4th horas? (b) Which rules the 5th? (c) Show that the 25th hora (the first of the next day) is ruled by Jupiter — so the next day is Thursday.

E3. [4] (a) Convert 7 ghatis 30 palas into hours and minutes. (b) Convert 5 h 36 min into ghatis. (c) How many muhurtas are in a 12-hour daytime? (d) How many ghatis make one prahar?

E4. [3] (a) By how many days do 12 lunar months fall short of a solar year? (b) After how many years does this shortfall add up to one lunar month (29.53 d)? (c) Convert to months — does it match "about every 32½ months"?

E5. [2] Calculate 19 × 365.2422 and 235 × 29.5306. How many **hours** apart are they?

E6. [3] (a) Which Hijri year will be running in December 2030? (b) In roughly which Gregorian year will 1500 AH begin? (c) Someone turns 66 by the Gregorian calendar — about how old are they in Hijri years?

E7. [3] (a) Which are leap years: 1600, 1800, 2000, 2026, 2100, 2400? (b) How many leap years were there from 1901 to 2000 inclusive?

E8. [2] 10 March 2027 is a Wednesday. Without a calendar, find the weekday of 10 March 2028 and of 10 March 2029.

E9. [3] (a) Ahmedabad is at 72.6° E. At what IST is its local noon (ignore the equation of time)? (b) Guwahati is at 91.7° E — when is its local noon? (c) Roughly how much earlier does the Sun rise in Guwahati than in Ahmedabad?

Section F — Calendar detective (project) ★★ 10 marks

Choose **any one date this month** and, using a panchang, a calendar app or a reliable website, write it in every calendar below.

Calendar	The date	Source used
Gregorian		
Indian National (Saka)		
Hindu lunisolar (month, paksha, tithi; Amanta or Purnimanta?)		
Nakshatra of the Moon that day		
Vikram Samvat year		
Hijri (day, month, year AH)		
Weekday in English, Hindi/Sanskrit and Arabic		

What phase was the Moon in on that date? Does it agree with the tithi? (2 lines)

Bonus challenges ★★★ up to 6 marks

X1. (3) The Gregorian calendar repeats exactly every 400 years, weekdays included. Show why: find the number of days in 400 years and divide by 7.

X2. (3) Ramadan began on about 8 February 2027. Using a drift of 10.88 days per year, in about which year will Ramadan first begin in July?